

# Differentiated Effects of Agent Diversity on Collective Decision-Making in LLM-Based Multi-Agent Delphi Systems

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## 1. RESEARCH OBJECTIVE & SYSTEM ARCHITECTURE

LLM-based Multi-Agent Delphi System (MADS). Operationalizes diversity across three dimensions: Variety (Big Five personality traits), Separation (Competing Values Framework belief archetypes), and Disparity (influence-weight distribution). Evaluates 11 group conditions across 4 open-weight LLMs.

## 2. BENCHMARK DATASETS & EMPIRICAL EVALUATION

Human-expert Delphi benchmark (8-person panel) on talent acquisition as external reference; agent-generated multi-round transcripts.

### Key Numerical Findings & Empirical Proofs:

Maximum variety yields highest exploration breadth but worst consensus efficiency and cohesion. Homogeneous groups achieve fast consensus but suffer from shallow agreement. Separation diversity (CVF opposing belief quadrants) is Pareto-optimal, balancing exploration breadth with high consensus convergence.

## 3. CRITICAL ARCHITECTURAL FLAWS & METHODOLOGICAL LIMITATIONS

### Identified System Failure Modes:

Unconditional 3-round execution on 100% of tasks; zero token cost optimization; high personality variety creates discursive friction without a dynamic filter; lacks primary agent confidence self-assessment.

## 4. RESEARCH GAP & RELEVANCE TO CAG-DELPHI

This work exemplifies the *Unconditional Consultation Dilemma* observed across the 15-paper corpus. By failing to incorporate epistemic uncertainty self-gating prior to multi-agent activation, the architecture incurs severe token inflation and exposes confident primary solutions to debate degeneration noise. In CAG-Delphi, this paper serves as an empirical anchor for diversity-governed consensus and comparative baseline evaluation.

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